Enterprise Sales Analytics Platform

Snowflake, Python and Power BI

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Contents

1 Project Overview

The objective of this project was to design and implement an end-to-end analytics platform capable of integrating CRM and ERP data into a centralized Snowflake data warehouse.

The solution follows the Medallion Architecture:

$$\text{Bronze} \rightarrow \text{Silver} \rightarrow \text{Gold}$$

The platform supports:

- Enterprise Reporting
- Customer Analytics
- Sales Analysis
- Forecasting Preparation
- Executive Dashboards

2 Business Problem

Organizations often store operational data across multiple systems, making it difficult to generate a unified view of business performance.

This project integrates:

2.1 CRM Data

- Customer Information
- Product Information
- Sales Transactions

2.2 ERP Data

- Customer Demographics
- Product Categories
- Location Information

The objective was to establish a single source of truth capable of supporting analytics, reporting, and future machine learning initiatives.

3 Snowflake Data Warehouse Architecture

3.1 Bronze Layer

The Bronze Layer stores source data exactly as received.

3.1.1 CRM Tables

- CRM_CUSTOMERS_RAW
- CRM_PRODUCTS_RAW
- CRM_SALES_RAW

3.1.2 ERP Tables

- ERP_CUSTOMERS_RAW
- ERP_PRODUCTS_RAW
- ERP_LOCATIONS_RAW

Responsibilities include:

- Data Ingestion
- Historical Retention
- Source System Auditability

3.2 Silver Layer

The Silver Layer provides standardized, cleaned, and integrated data.

3.2.1 Customer Transformations

- DIM_CUSTOMER_STG
- ERP_CUSTOMER_STG

Activities:

- Gender Standardization
- Marital Status Standardization
- Data Cleansing
- Customer Integration

3.2.2 Product Transformations

- DIM_PRODUCT_STG
- ERP_PRODUCT_STG

Activities:

- Product Standardization
- Product Enrichment
- Data Quality Validation

3.2.3 Sales Transformations

- FACT_SALES_STG

Activities:

- Date Transformations
- Revenue Standardization
- Sales Cleansing

3.3 Gold Layer

The Gold Layer provides reporting-ready business datasets.

3.3.1 Dimension Tables

- DIM_CUSTOMER
- DIM_CUSTOMER_CLEAN
- DIM_PRODUCT
- DIM_PRODUCT_PBI_CLEAN

3.3.2 Fact Tables

- FACT_SALES

4 Data Model

A Star Schema was implemented.

```
DIM_CUSTOMER_CLEAN
         |
         |
         |
    FACT_SALES
         |
         |
         |
DIM_PRODUCT_PBI_CLEAN
```

4.1 Fact Table

FACT_SALES contains:

- Order Number
- Customer ID
- Product Key
- Order Date
- Quantity
- Unit Price
- Sales Amount

4.2 Dimension Tables

DIM_CUSTOMER_CLEAN contains:

- Customer Information
- Gender
- Marital Status

DIM_PRODUCT_PBI_CLEAN contains:

- Product Information
- Product Attributes

5 Data Quality Challenges

5.1 Customer Dimension Issues

Challenges encountered:

- Duplicate Customer IDs
- Null Customer IDs

Solution:

- Created DIM_CUSTOMER_CLEAN
- Applied ROW_NUMBER() based deduplication
- Removed null customer records

5.2 Product Dimension Issues

Challenges:

- Duplicate Product Keys

Solution:

- Created DIM_PRODUCT_PBI_CLEAN
- Removed duplicate product records

5.3 Product Key Mapping Issue

A mismatch was identified between:

- FACT_SALES.PRODUCT_KEY
- DIM_PRODUCT.PRODUCT_KEY

The issue was documented and isolated for future remediation.

6 Python Analytics Layer

The Snowflake Gold Layer was connected to Jupyter Notebook using Python.

6.1 Technologies

- Snowflake Connector
- Pandas
- NumPy
- Matplotlib

6.2 Data Extraction

The following tables were imported into Pandas DataFrames:

- FACT_SALES
- DIM_CUSTOMER
- DIM_PRODUCT

6.3 Data Cleaning

Activities included:

- Date Conversion
- Missing Value Handling
- Data Type Validation

7 Exploratory Data Analysis

7.1 Monthly Sales Trend

Business Question:

How has sales performance evolved over time?

Outcome:

- Revenue Trend Visualization
- Time-Series Analysis

7.2 Customer Analysis

Business Question:

Which customers generate the highest revenue?

Outcome:

- Customer Revenue Ranking
- Customer Contribution Analysis

8 Power BI Analytics Layer

The Snowflake Gold Layer was connected directly to Power BI.

8.1 Power BI Data Model

- DIM_CUSTOMER_CLEAN
- DIM_PRODUCT_PBI_CLEAN
- FACT_SALES

8.2 Relationships

Customer Relationship:

```
DIM_CUSTOMER_CLEAN[CUSTOMER_ID]
                  |
                  |
FACT_SALES[CUSTOMER_ID]
```

Product Relationship:

```
DIM_PRODUCT_PBI_CLEAN[PRODUCT_KEY]
                  |
                  |
FACT_SALES[PRODUCT_KEY]
```

9 Sales Executive Dashboard

An executive dashboard was developed to support decision-making.

9.1 Key Performance Indicators

1. Total Revenue: $29M
2. Total Orders: 28K
3. Total Customers: 18K
4. Total Quantity Sold: 60.42K
5. Average Order Value: $1.06K

9.2 Dashboard Components

- Revenue Trend Analysis
- Top Customers by Revenue
- Revenue by Gender
- Revenue by Marital Status
- Interactive Filters

9.3 Filters

- Order Date
- Gender
- Marital Status

10 Key Business Insights

10.1 Revenue Performance

The organization generated approximately $29 million in revenue.

10.2 Customer Base

The organization serves approximately 18,000 unique customers.

10.3 Revenue Growth

Monthly sales trends indicate strong growth throughout the observed period.

10.4 Customer Segmentation

Analysis revealed differences in revenue contribution across:

- Gender Segments
- Marital Status Groups
- High-Value Customers

11 Future Enhancements

11.1 Machine Learning

Implement:

- Sales Forecasting
- Demand Forecasting
- Customer Segmentation Models

Algorithms:

- Linear Regression
- Random Forest
- XGBoost

11.2 Forecast Dashboard

Develop:

- Actual vs Forecast Revenue
- Forecast Trend Analysis
- Revenue Projections

12 Conclusion

This project demonstrates a complete modern analytics architecture from data ingestion through executive reporting.

Achievements include:

- Snowflake Data Warehouse Implementation
- Bronze-Silver-Gold Architecture
- Data Quality Management

- Python Analytics Layer
- Exploratory Data Analysis
- Power BI Executive Dashboard
- Customer Analytics
- Star Schema Data Modeling

The solution establishes a scalable foundation for future machine learning, forecasting, and advanced analytics initiatives.